HARSH GUPTA

VIT BHOPAL UNIVERSITY, SEHORE • 91+7877454334 • hkgupta160420@gmail.com

Linkedin • Github

SUMMARY

Third-year B.Tech Computer Science student focused on Artificial Intelligence, Machine Learning, and modern generative AI systems, including optimization and deployment of LLaMA-based models. Skilled in Python, Java, and C++, with hands-on experience in deep learning pipelines, full-stack engineering, and scalable cloud-native architectures. Strong foundation in computational linguistics and applied AI, enabling the development of production-ready solutions across LLMs, RAG workflows, and computer vision for real-world, high-impact applications.

EDUCATION

- **Vellore Institute of Technology, Bhopal** — Sep 2023 - Present
  Bachelor of Technology in Computer Science (Artificial Intelligence & Machine Learning)
  Current CGPA: 8.99/10.0 (Top 5% of cohort)
- Secondary School Examination, CBSE, 2020 — 80% — 2009 - 2020
- Senior School Certificate Examination, CBSE, 2022 — 85% — 2020 - 2022

PROJECTS

VAIDYASETU

- Built VaidyaSetu – AI healthcare prototype with telemedicine (WebRTC), OCR–LLM, AI diagnostics, improving accessibility +35% (20 users).
- Engineered OCR–LLM + Google Calendar API for report extraction & auto-scheduling, cutting effort ~60%.
- Trained CNN (10K+ images, 96% acc.) and integrated NLP chatbot, boosting report comprehension +40%.

LLAMA-3 QLORA FINE-TUNING FOR HEART DISEASE RECOMMENDATIONS

- Fine-tuned LLaMA-3 8B with QLoRA + LoRA, reducing GPU memory 50% while maintaining accuracy.
- Automated preprocessing & publishing of 2K+ JSON → Hugging Face datasets, cutting manual effort 70%.
- Optimized training (transformers, TRL, accelerate, bitsandbytes) with grad checkpointing + AdamW-8bit, improving efficiency 30% and reproducibility.

ADVANCED BRAIN CANCER DETECTION USING CNN ARCHITECTURES

- Implemented ensemble CNN models using VGG16, MobileNet, and ResNet50 achieving 99.8% classification accuracy on BraTS2020 medical dataset
- Preprocessed 1,000+ MRI images using OpenCV libraries and deployed intuitive clinical interface with Flask containerization technology
- Authored 7-page reviewed research paper documenting comprehensive methodology under faculty supervision

EXPERIENCE

AI/ML Intern — EnviroWealth (Dec 2025 – Feb 2026)

- Engineered an AI-powered BRSR ESG automation pipeline using Python, FastAPI, Gemini LLM, and Pandas to extract and structure 400+ compliance fields from PDF/Excel documents via async REST APIs, reducing manual reporting effort by ~70%.
- Designed a 3-layer NLP transformation architecture with chunked Gemini API prompting across 8 document sections, custom flat-to-nested JSON parser in Python, MongoDB data storage, and Dockerized full-stack deployment on Vercel.

ACHIEVEMENTS

- Winner of VIT Mother’s Day Challenge, designing an innovative solution that strengthened engagement and awareness among 100+ participants.
- Led team to 3rd place in Hack4Health Hackathon at VIT

PUBLICATIONS / RESEARCH PAPERS

- Diabetes Risk Prediction with Smart Healthcare Model Based on Machine Learning (IEEE ICTBIG 2025, SUAS Indore)
- Improving Intrusion Detection Using Multiclass Cyberattack Classification with LightGBM, XGBoost, and Random Forest (IEEE ICTBIG 2025, SUAS Indore)

SKILLS AND CERTIFICATIONS

Technical Skills:

- AI/ML: Scikit-learn, Pandas, NumPy, TensorFlow, RAG, Generative AI, LLMs, CNNs, Effi cientNet, XGBoost, LightGBM
- Programming: Java (Data Structures & Algorithms), C++, Python, HTML/CSS, MySQL
- Cloud, DevOps & Tools: Docker, Render, Git, GitHub, FastAPI, Flask, Streamlit, Postman, VS Code

Certifications:

- AWS Certified Machine Learning Engineer - Associate – July’25 (link)
- GEN AI using IBM Watsonx by IBM Developer Skill Network – June’25 (link)
- MATLAB and Simulink Technical Computing Certification
- GeeksforGeeks 160-Day DSA Mastery Certificate